

TEST YOURSELF: TECHNICAL REPORTING AND EVIDENCE ANALYSIS



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Project: Lab 03 - Metasploitable 2 Reconnaissance

Subject: IT Security and Privacy

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Date: 2026-02-10

Location: Bogotá, Colombia

Confidentiality: Internal / Academic Use Only



1. Structure: Report Sections and Differences

1.1 Essential Sections in a Pentest Technical Report

A professional technical report must contain, at a minimum, the following sections to be actionable:

- ♦ **Executive Summary:** A high-level overview for management.
- ♦ **Scope & Methodology:** Testing boundaries and techniques used.
- ♦ **Vulnerability Summary Table:** A quick inventory of findings categorized by severity.
- ♦ **Detailed Findings:** Technical description, reproduction steps, impact, and evidence for each vulnerability.
- ♦ **Remediation & Recommendations:** Specific steps to mitigate the identified risks.
- ♦ **Appendices/Logs:** Raw tool outputs (e.g., Nmap scans) that support the analysis.

1.2 Difference: Technical vs. Executive Report

The key difference lies in the **audience and purpose**:

- ♦ **Executive Report:** Designed for strategic decision-making. It avoids excessive technical jargon and focuses on **business risk**, the budget required for fixes, and the impact on operational continuity.
- ♦ **Technical Report:** Designed for system administrators and developers. It focuses on the **"how"**, providing exact commands, software versions, traffic captures, and precise details that allow the IT team to replicate and patch the vulnerability.



2. Evidence: Integrity and Verification

2.1 Importance of SHA256 Hashing

Calculating the SHA256 hash is vital to ensure **evidence integrity** and **non-repudiation**. In a forensic or legal audit context, the hash acts as a unique digital fingerprint. If a client or third party questions the veracity of the data, the hash proves that the file presented is exactly the same as the one generated during the test.

2.2 Consequences of Altered Evidence

If the evidence were altered (even by a single bit), the SHA256 hash would change drastically. This would result in:

1. **Loss of Credibility:** The report would lose all professional validity before a client or a court of law.
2. **Technical Inaccuracy:** Recommendations could be based on false data, leading to ineffective remediations.
3. **Legal Issues:** In compliance audits (such as PCI-DSS), altering logs or reports can lead to severe penalties and legal liabilities.



3. Practice: Finding Entry - Port 21 (Anonymous FTP)

Finding: Unrestricted Anonymous FTP Access

Summary:

The FTP service allows users to log in with the "anonymous" account without providing a password. This configuration permits unauthenticated users to browse or potentially upload/download files from the server.

Technical Description:

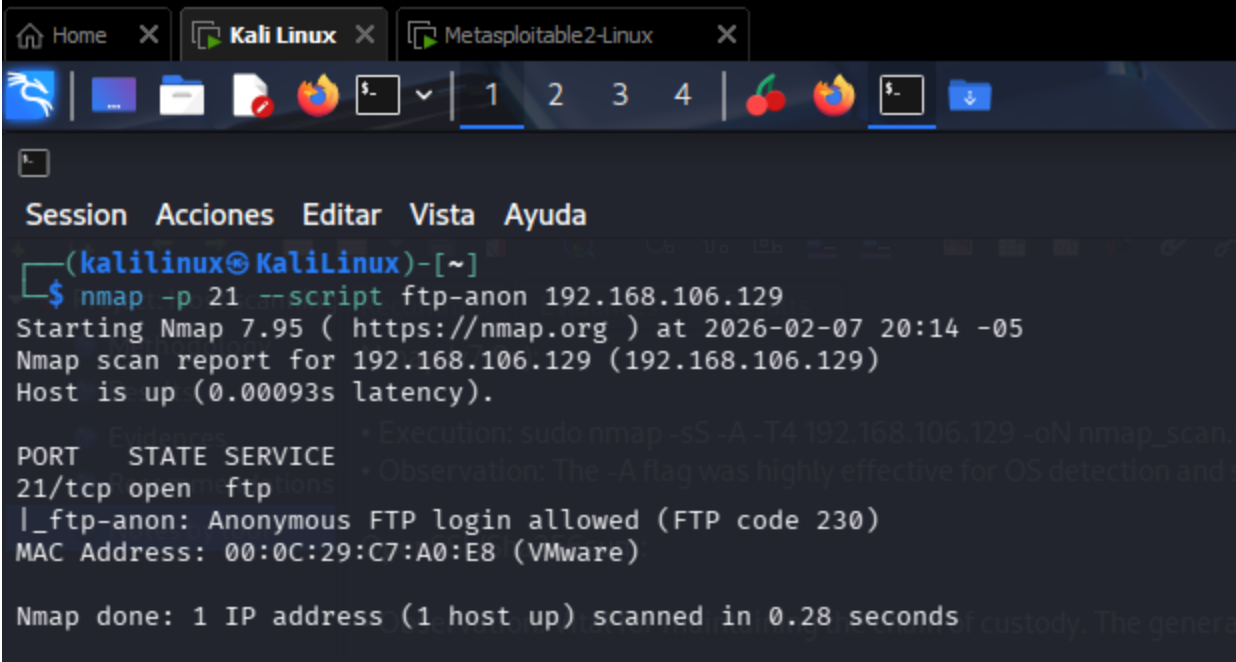
During the scanning phase, the Nmap script `ftp-anon` detected that the FTP service on port 21 permits anonymous logins. This is often caused by a misconfiguration in the `vsftpd.conf` file (specifically the parameter `anonymous_enable=YES`).

Impact:

- ◆ **Information Disclosure:** Attackers can steal sensitive data hosted on the FTP share.
- ◆ **Service Enumeration:** Attackers gain insights into the file structure.
- ◆ **Note on Write Permissions:** Manual testing confirmed that while read access is granted, write permissions are restricted, mitigating the risk of unauthorized file uploads.

Attached Evidence:

Capture 1: Vulnerability Detection



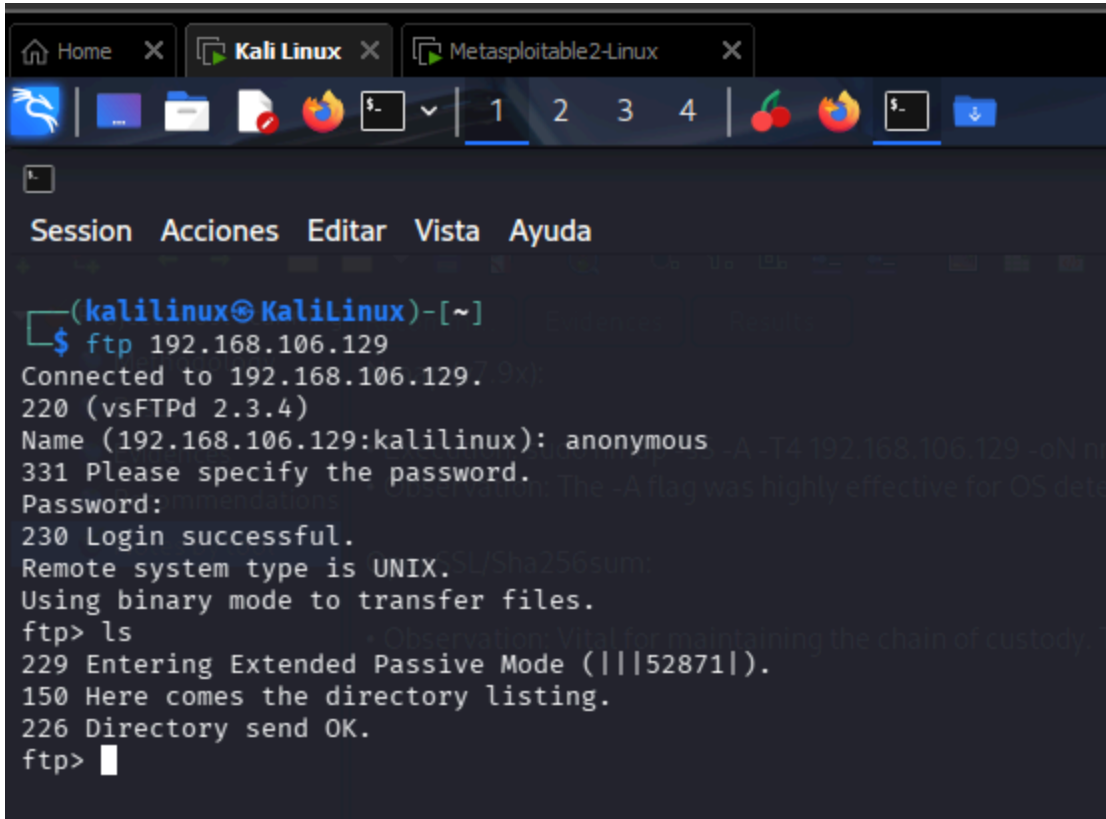
```
Session Acciones Editar Vista Ayuda
(kalilinux@KaliLinux)-[~]
$ nmap -p 21 --script ftp-anon 192.168.106.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-07 20:14 -05
Nmap scan report for 192.168.106.129 (192.168.106.129)
Host is up (0.00093s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
MAC Address: 00:0C:29:C7:A0:E8 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds
```

Figure 1: Nmap output confirming that the service vsFTPD 2.3.4 allows anonymous login (Code 230).

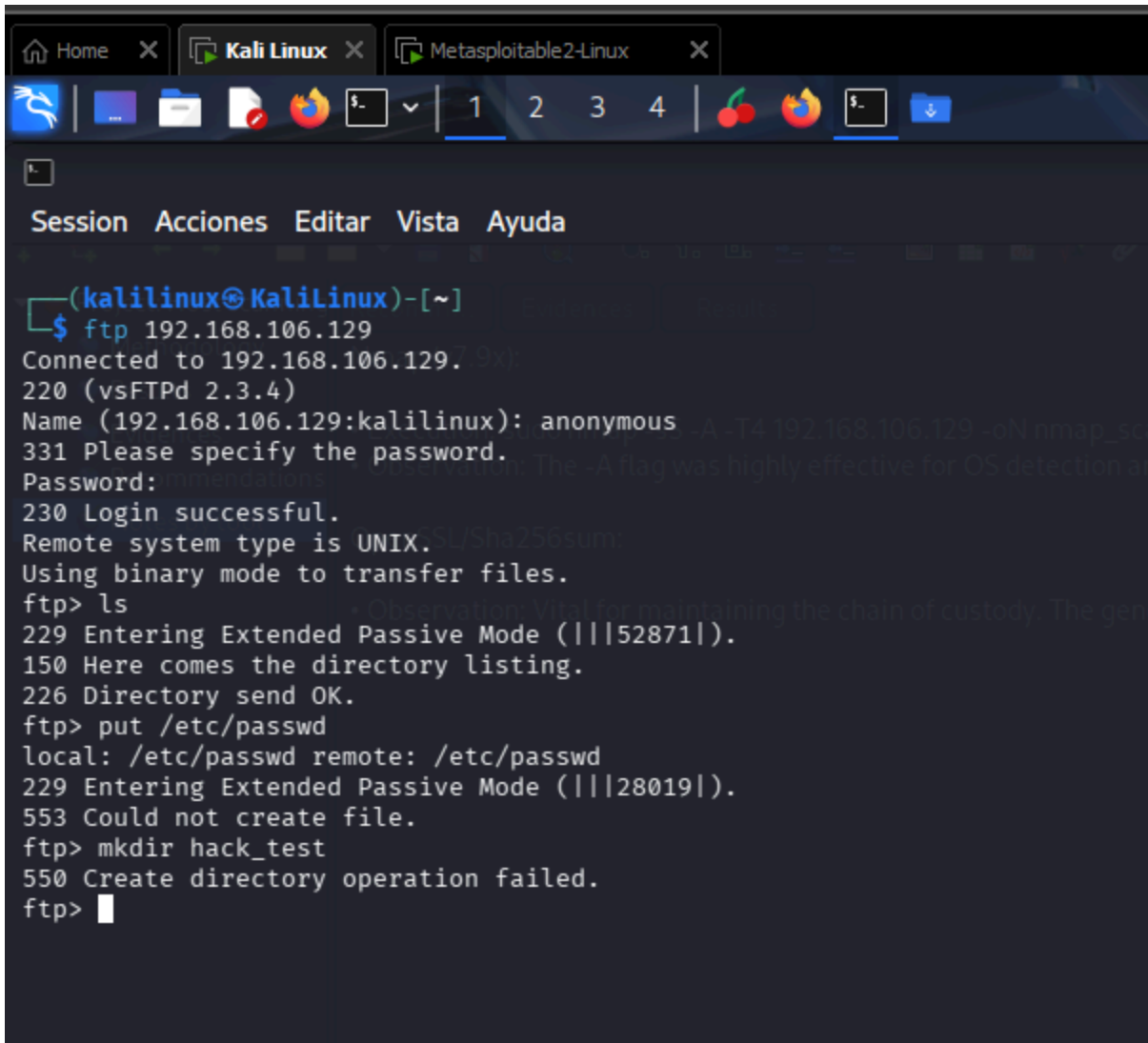
Capture 2: Successful Exploitation (Information Disclosure)



```
(kalilinux@KaliLinux)-[~]  
$ ftp 192.168.106.129  
Connected to 192.168.106.129.  
220 (vsFTPd 2.3.4)  
Name (192.168.106.129:kalilinux): anonymous  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> ls  
229 Entering Extended Passive Mode (|||52871|).  
150 Here comes the directory listing.  
226 Directory send OK.  
ftp> 
```

Figure 2: Manual connection from Kali Linux using 'anonymous' credentials, providing a full directory listing (ls).

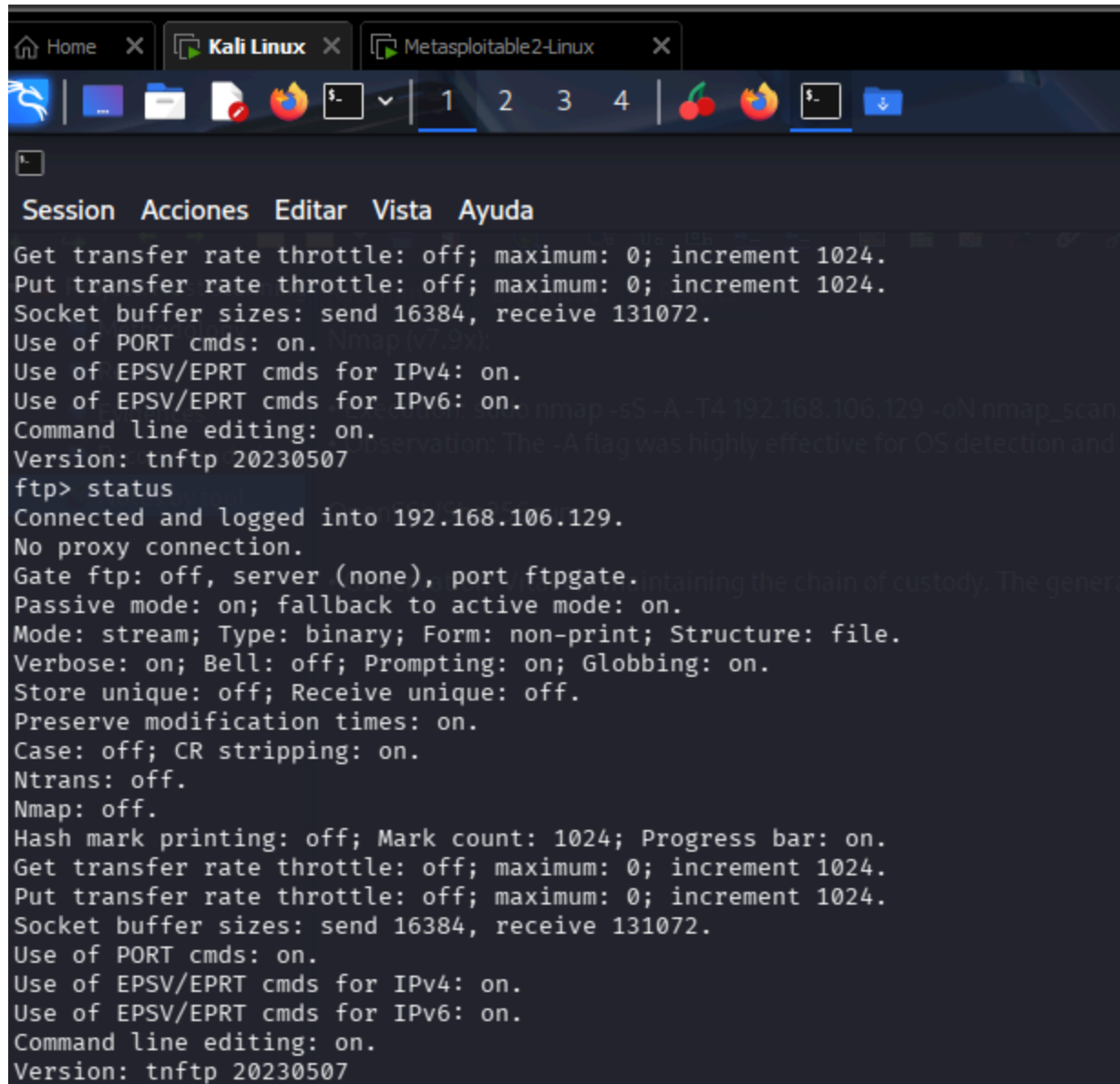
Capture 3: Write Permission Test (Impact Analysis)



```
(kalilinux@KaliLinux)-[~]  
$ ftp 192.168.106.129  
Connected to 192.168.106.129.  
220 (vsFTPD 2.3.4)  
Name (192.168.106.129:kalilinux): anonymous  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> ls  
229 Entering Extended Passive Mode (|||52871|).  
150 Here comes the directory listing.  
226 Directory send OK.  
ftp> put /etc/passwd  
local: /etc/passwd remote: /etc/passwd  
229 Entering Extended Passive Mode (|||28019|).  
553 Could not create file.  
ftp> mkdir hack_test  
550 Create directory operation failed.  
ftp>
```

Figure 3: Attempting to upload a file (put) and create a directory (mkdir). The 'Permission denied' errors confirm that the impact is limited to unauthorized data access.

Capture 4: Connection Status



```
Session Acciones Editar Vista Ayuda
Get transfer rate throttle: off; maximum: 0; increment 1024.
Put transfer rate throttle: off; maximum: 0; increment 1024.
Socket buffer sizes: send 16384, receive 131072.
Use of PORT cmds: on.
Use of EPSV/EPRT cmds for IPv4: on.
Use of EPSV/EPRT cmds for IPv6: on.
Command line editing: on.
Version: tnftp 20230507
ftp> status
Connected and logged into 192.168.106.129.
No proxy connection.
Gate ftp: off, server (none), port ftpgate.
Passive mode: on; fallback to active mode: on.
Mode: stream; Type: binary; Form: non-print; Structure: file.
Verbose: on; Bell: off; Prompting: on; Globbing: on.
Store unique: off; Receive unique: off.
Preserve modification times: on.
Case: off; CR stripping: on.
Ntrans: off.
Nmap: off.
Hash mark printing: off; Mark count: 1024; Progress bar: on.
Get transfer rate throttle: off; maximum: 0; increment 1024.
Put transfer rate throttle: off; maximum: 0; increment 1024.
Socket buffer sizes: send 16384, receive 131072.
Use of PORT cmds: on.
Use of EPSV/EPRT cmds for IPv4: on.
Use of EPSV/EPRT cmds for IPv6: on.
Command line editing: on.
Version: tnftp 20230507
```

Figure 4: Detailed connection status showing the 'tnftp' client configuration and active session parameters.

Recommendation:

1. **Disable Anonymous Login:** Set `anonymous_enable=NO` in the FTP configuration.
2. **Implement Secure Protocols:** Transition from FTP to SFTP (Port 22) for encrypted transfers.

